

Income Inequality, Capitalism, and Diversity: A Goodness-of-fit Decomposition Perspective

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Abstract

This research explores the intricate dynamics of income redistribution in the context of economic freedom, diversity, and various socio-economic factors. Through the innovative lens of goodness-of-fit decomposition, we analyze the individual contributions of economic freedom, predicted diversity, religious composition, economic indicators, continent-specific factors, and legal origins in explaining income inequality. Our comprehensive dataset, spanning 1971-2010 and encompassing 108 countries, draws from the Ashraf and Galor dataset (2013) and the Sturm and De Haan dataset (2015), providing a robust foundation for our analysis. The findings emphasize the critical role of balancing economic freedom and diversity in achieving equitable income distribution. Additionally, certain legal origins, notably Scandinavia, exert substantial influence, while religious composition, particularly the presence of Protestants and Muslims, shapes income inequality. These insights are pivotal for informing policies addressing income inequality, considering regional disparities and cultural contexts.

Keywords: income redistribution, economic freedom, predicted diversity, religious composition, economic indicators, continent factors, legal origins

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Table of Contents

1. Introduction	3
Previous Findings.....	3
Contribution to Research.....	4
2. Methodology	5
3. Data Description.....	6
4. Econometric Model	7
Income Redistribution and Economic Freedom, Predicted Diversity, Including Religious Composition, Economic Indicators, Continent Dummies, and Legal Origins.....	7
5. Application of Goodness-of-fit Decomposition	10
“Basic” model	10
Log GDP per capita.....	12
KOF economic globalization index.....	13
Continent-Specific Factors	14
Legal Origins.....	15
Religious Composition.....	17
6. Combination of Factors	20
“Basic” model, Including Economic Indicators and Continent Dummies	21
“Basic” model, Including Economic Indicators and Legal Origins	21
“Basic” model, Including Economic Indicators and Religious Composition	22
7. Conclusion.....	23
References	27
Appendix	28

1. Introduction

Income inequality is a pervasive and pressing issue that continues to shape societies worldwide. Within the framework of economic models and sociopolitical debates, scholars have sought to unravel the complex web of factors that contribute to this inequality. In this study, we delve into the realms of income inequality, capitalism, and diversity, offering a unique perspective through the lens of goodness-of-fit decomposition. Our primary aim is to dissect the explanatory power of various factors commonly associated with income redistribution and shed light on their individual contributions to the overall understanding of this critical socioeconomic concern. Through rigorous analysis and empirical investigation, we seek to unveil the nuanced interplay between economic freedom, diversity, legal origins, religious composition, and globalization, providing valuable insights for policymakers and researchers alike.

Previous Findings

The relationship between income inequality, capitalism, and diversity has been a subject of extensive research, drawing upon various theoretical frameworks and empirical approaches. In this literature review, we examine key studies and concepts that are pertinent to our investigation of income inequality, capitalism, and diversity from a goodness-of-fit decomposition perspective.

Desmet, Ortuño-Ortín, and Weber (2009) investigate the influence of linguistic diversity on income redistribution. Their findings indicate that linguistic diversity holds both statistical and economic significance in understanding redistribution. Linguistic distances, which capture the dissimilarity between languages, provide a more nuanced perspective on this relationship. Specifically, an increase in linguistic diversity correlates with a decrease in income redistribution, highlighting the complex interplay between diversity and redistribution levels. This study underscores the importance of considering diverse linguistic backgrounds when analyzing income inequality and redistribution (Desmet et al., 2009).

Sturm and de Haan (2015) explore the relationship between economic freedom, ethno-linguistic fractionalization, and income inequality. They find that economic freedom, proxied by an adjusted economic freedom index, does not consistently relate to income inequality across a broad sample of countries. Importantly, they note that the impact of ethno-linguistic fractionalization on income redistribution is contingent upon the level of economic freedom. In countries with high fractionalization, income redistribution tends to be lower, while capitalist nations with low fractionalization exhibit greater income redistribution. This study highlights the interaction between economic freedom, diversity, and income inequality (Sturm & de Haan, 2015).

Ashraf and Galor (2013) delve into the historical aspect of diversity and its impact on economic development. They examine the influence of genetic diversity resulting from ancient migratory patterns on comparative economic development. The study suggests that intermediate levels of genetic diversity have been beneficial to Asian and European populations, while high genetic diversity in Africa and low diversity in Native American populations have posed challenges to development. This research broadens the diversity perspective beyond linguistic or ethno-linguistic factors, emphasizing the relevance of genetic diversity in understanding economic outcomes (Ashraf & Galor, 2013).

More recently, Ibrahim Ben Araar (2022) revisited the relationship between diversity, economic freedom, and income redistribution. The study finds a negative linear effect of genetic diversity on redistribution. Importantly, it highlights the influence of economic freedom and ethnic fractionalization on redistribution, reinforcing the significance of these factors in shaping income inequality. The study supports the importance of a capitalist system and genetic diversity for higher levels of income redistribution while questioning the suitability of genetic diversity as a standalone variable (Araar, 2022).

In sum, these key studies provide valuable insights into the multifaceted relationship between income inequality, capitalism, and diversity. They underscore the importance of considering linguistic, ethno-linguistic, and genetic diversity, as well as economic freedom, when examining income redistribution.

Contribution to Research

This study significantly advances our understanding of income inequality, capitalism, and diversity by extending the analysis of a merged dataset from Ashraf and Galor (2013) and Sturm and De Haan (2015) proceeded by Ibrahim Ben Araar. A key innovation lies in the use of the “rego method” to decompose the R-squared goodness-of-fit measure of an OLS regression model. This approach offers a more nuanced examination of the relationships between these factors. By dissecting the goodness-of-fit measure, the study aims to uncover hidden patterns and contributions of individual factors to the model. Furthermore, this extension of Araar's analysis provides a fresh perspective on the genetic diversity-redistribution relationship. Ultimately, this research contributes valuable insights to inform policy decisions and deepen our understanding of these societal dynamics.

Relevance of the Study

The relevance of this study cannot be overstated, particularly in an era marked by increasing disparities in wealth and income. Income inequality has far-reaching implications for societal

cohesion, economic stability, and overall well-being. By dissecting the intricate relationships between capitalism, diversity, and income inequality, we provide valuable insights that can inform both policy formulation and future research directions. Moreover, by examining the explanatory power of specific factors, we contribute to the broader conversation on income redistribution, enhancing our understanding of the mechanisms at play in contemporary economies.

Research Objectives:

1. To dissect and quantify the individual explanatory power of various factors, including economic freedom, globalization, religious composition, geographical location and legal origins, in explaining income inequality.
2. To explore how the interactions between economic variables, diversity measures, and geographic regions influence income redistribution.
3. To provide insights into the implications of our findings for public policies aimed at reducing income inequality.

Research Questions:

1. What is the relative contribution of economic freedom, diversity, and other key variables to the explanation of income inequality?
2. How do interactions between these factors influence income redistribution patterns?

2. Methodology

Goodness-of-fit tests are essential in assessing how well a regression model aligns with observed data, essentially measuring how accurately the model explains variations in the dependent variable. At its core, this assessment is pivotal for evaluating the model's suitability and reliability.

R-squared Goodness-of-fit Measure is often referred to as the coefficient of determination. R-squared plays a central role in quantifying the strength of the relationship between the model and the dependent variable. It operates on a scale from 0% to 100%, where a higher R-squared value signifies that the independent variable(s) within the model account for a substantial proportion of the variations in the dependent variable. Conversely, a lower R-squared value indicates that the independent variable(s) have limited explanatory power.

Decomposition with the Rego Method, implemented as a Stata module developed and maintained by Frank Huettner and Marco Sunder at the University of Leipzig, is a valuable tool for dissecting the R-squared goodness-of-fit measure (Sunder, M., 2011). It provides insights into

how individual independent variables contribute to the overall model fit, using the principles of cooperative game theory.

Decomposition involves assessing each independent variable's contribution to the model's R-squared value. The Shapley value method calculates each variable's average contribution across all potential combinations, ensuring a fair distribution of total explanatory power. Conversely, the Owen value method assigns contributions based on independent variables' membership in predefined groups.

In cooperative game theory, the **Shapley value** method ensures equitable distribution of the total coalition surplus. It adheres to efficiency, symmetry, additivity, and null player principles. The **Owen value** introduces a "reference player" into each coalition, distributing their contribution based on a specific rule. It involves a two-step process, with unions playing a quotient game, and the payoff shared among players using the Shapley value.

The Rego Method aligns with Shapley and Owen values' principles, proving beneficial for grouping similar variables for analysis.

The relevance of the Rego method to our research lies in its ability to provide a quantitative measure of how well our model captures the dynamics between income inequality, capitalism, and diversity. By decomposing the R-squared value, we can identify the unique contributions of individual independent variables to the overall model fit. This approach allows us to assess which factors have a more substantial impact on explaining the variation in income inequality within the context of capitalism and diversity.

3. Data Description

This research relies on a merged dataset, combining data from two primary sources: the Ashraf and Galor dataset (2013) and the Sturm and De Haan dataset (2015).

The Ashraf and Galor dataset, spanning 108 countries across 8 five-year intervals from 1971 to 2010, is rooted in genetic diversity projections derived from the Human Genome Diversity Cell Line Panel (Cann, H. M., 2002). Genetic diversity predictions are based on the "Out-Of-Africa-Hypothesis," which posits that humanity's origins trace back to East Africa, specifically Ethiopia. To predict genetic diversity, measures consider the distance from Ethiopia and migratory waypoints. Genetic diversity is quantified using expected heterozygosity and includes an ancestry-adjusted predicted genetic diversity variable. Additionally, squared variables are employed to examine hump-shaped relationships within the data. However, genetic diversity data may exhibit false correlations, especially when neighboring countries to Ethiopia are considered. The

relationship between genetic diversity, interpersonal trust, and cooperation requires critical examination.

The Sturm and De Haan dataset covers 108 countries over the same period, 1971-2010. This dataset includes a dependent redistribution measure sourced from the Standardized World Income Inequality Database (SWIID) (Solt, F., 2016) and the Economic Freedom Index from the Fraser Institute (Lawson, R. et al., 2015), a measure used to gauge capitalism. To mitigate potential endogeneity issues, the economic freedom variable is presented as a five-year average for each observation. It is important to note that the Fraser Institute's economic freedom index may bear a pro-capitalist bias.

Ibrahim Ben Araar merged these datasets in 2022, resulting in a comprehensive dataset with 23,707 observations. This rich dataset serves as the foundation for our analysis of income inequality, capitalism, and diversity. It provides a robust basis for our research while acknowledging potential sources of bias and the need for critical examination, especially in the context of genetic diversity.

4. Econometric Model

Let us commence our exploration by delving into the primary discoveries of Ibrahim Ben Araar's (2022) econometric study, which leveraged the merged database discussed earlier. These findings will serve as a valuable foundation for our subsequent analysis of R-squared goodness-of-fit decomposition.

Income Redistribution and Economic Freedom, Predicted Diversity, Including Religious Composition, Economic Indicators, Continent Dummies, and Legal Origins

Araar's investigation, as outlined in Table 2, examines the crucial factors that wield significant influence over income redistribution. To gain a comprehensive understanding of these dynamics, the study takes into account a wide array of control variables. These variables encompass not only the fundamental Economic Freedom and Predicted Diversity indexes but also include religious composition, economic indicators like GDP per capita and the globalization index, continent-specific factors, and legal origins. The study's outcomes, which we will scrutinize using the Goodness-of-fit Decomposition method, involve a series of ten linear regression models arranged in a panel format, as summarized in Table 2.

	1	2	3	4	5	6	7	8	9	10
Economic Freedom	-0.423*** (-7.90)	-0.361* (-1.77)	-0.314*** (-6.17)	-0.428*** (-7.80)	-0.214*** (-4.00)	-0.228*** (-5.13)	-0.361*** (-6.74)	-0.407*** (-8.38)	-0.398*** (-7.12)	-0.353*** (-6.62)

Predicted Diversity	-3.168*** (-6.78)	-3.02 (-1.63)	-1.558*** (-3.41)	-3.244*** (-6.72)	-2.473*** (-4.50)	-1.697*** (-4.40)	-2.343*** (-4.90)	-3.021*** (-7.13)	-2.865*** (-5.65)	-2.394*** (-5.09)
EF x Predicted Diversity	0.733*** -9.64	0.667** -2.34	0.488*** -6.59	0.744*** -9.44	0.357*** -4.690	0.418*** -6.57	0.629*** -8.22	0.680*** -9.84	0.697*** -8.76	0.632*** -8.34
Log GDP per capita			0.0944*** -13.08							
KOF globalization index				-0.000338 (-0.97)						
Europe					0.388*** -8.650					
Asia					0.0623* -1.660					
Africa					0.112** -2.340					
North America					0.239*** -6.670					
Oceania					0.138*** -4.510					
Socialist						0.101*** -7.15				
Germany						0.139*** -6.35				
Scandinavia						0.442*** -21.88				
Muslim							-0.00142*** (-6.09)			
Protestant								0.00319*** -14.29		
Catholic									0.000266 -1.53	

Other religions									-0.00128***	
									(-6.84)	
Constant	2.857***	2.611*	1.282***	2.913***	2.524***	1.943***	2.368***	2.840***	2.637***	2.359***
	-8.7	-1.97	-3.85	-8.56	-6.830	-7.1900	-7.1300	-9.53	-7.36	-7.17
Observations	943	100	918	928	943	943	943	935	943	935
R2	0.468	0.514	0.549	0.465	0.660	0.6510	0.4880	0.563	0.469	0.492

Notes: The dependent variable is income redistribution. The first column is our regular model, the second column represents the model for the year 2005. The other columns include a variety of dummy variables in our basic model. T – statistics in parentheses. Sources: (Sturm & De Haan, 2015a) and (Ashraf & Galor, 2013)

***Significant at the 1 percent level.
**Significant at the 5 percent level
*Significant at the 10 percent level.

Table 2. Model with all explored regressors.

The research findings unveil several discernible patterns:

Firstly, countries situated in North America, Europe, and Oceania tend to have higher levels of income redistribution, with Europe exerting the most significant impact among them. This observation aligns with the well-established reputation of European nations for their robust social security systems and higher taxation levels, contributing to more equitable income distribution.

Secondly, an increase in GDP per capita correlates with greater income redistribution, suggesting that wealthier societies are inclined to implement more extensive income redistribution mechanisms. This relationship emphasizes the role of economic prosperity in promoting economic equality.

Interestingly, the globalization index does not exhibit any significant influence on income redistribution in this analysis, indicating that trade openness may not play a substantial role in shaping redistribution dynamics.

Legal origin emerges as a crucial determinant in income redistribution. Scandinavian legal systems, renowned for their emphasis on social equality, exert the most significant influence in promoting income redistribution. This aligns with the reputation of Scandinavian countries as champions of economic equality.

Intriguingly, the German legal origin is found to contribute to more income redistribution than the socialist legal origin. This unexpected result highlights the need for a nuanced understanding of legal systems and their impact on economic policies.

In terms of religion, it plays a somewhat smaller role in income redistribution. Protestants are found to have a positive influence on redistribution efforts, indicating a greater willingness to

address income inequality. In contrast, the presence of Muslims and other religious groups tends to have a negative impact on income redistribution.

In summary, this analysis underscores the multifaceted nature of income redistribution, emphasizing the significance of legal origin, economic indicators, and regional factors. It underscores the importance of considering a diverse set of variables to gain a comprehensive understanding of the dynamics of income redistribution and its implications for economic equity on a global scale.

5. Application of Goodness-of-fit Decomposition

In our research, we applied the Rego method to dissect the R-squared goodness-of-fit measure of each of the ten Ordinary Least Squares (OLS) regression models in STATA, as discussed in the previous section. This method enabled us to discern the individual contributions of each regressor to the overall goodness-of-fit, facilitating meaningful conclusions based on the outcomes.

While we have previously discussed the impact of each factor, both in terms of its direction (positive or negative) and its magnitude, it is now pertinent to assess their explanatory value. In this section, we will examine how effectively certain factors can account for variations in income redistribution, essentially evaluating their significance. The Rego method will reveal the fractions of R-squared attributed to each variable within the context of the overall linear model. In essence, the Rego method will demonstrate the effectiveness of each factor in explaining the dependent variable, income redistribution, and consequently, their importance in our analysis. The results, displayed in tables under the "Shapley %R2" column, represent the outcomes of the Rego method. These percentages signify the contribution of each factor to the overall R-squared value. It is worth noting that across all models, R-squared ranges from 46% to 66%.

“Basic” model

Let us begin by examining the two "basic models" that form the foundation of our analysis. In these models, the dependent variable is income redistribution in a given country, and the independent variables consist of the Economic Freedom Index, Predicted Diversity, and the product of these two coefficients [Table 3 and 4].

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.4228996	***	0.0535461	0.000	-2.5533	37.4295
Predicted Diversity	-3.168471	***	0.4670933	0.000	-0.7631	13.0194
EF x Predicted Diversity	0.7330411	***	0.0760327	0.000	3.3500	49.5510
Constant	2.8573		0.3284286	0.000		

Observations	943		
Overall R2	0.46811		
Root MSE	0.1716714		
F-stat. Model	275.4713	***	0.000
Log Likelihood	325.6744		

Table 3. “Basic” model.

Specifically, Table 4 presents the model for the 2005 year. This model was introduced by Araar to facilitate comparisons with the findings of Sturm & De Haan (2015a). It is important to highlight that in the 2005 models, only the economic freedom variable and the interaction term prove to be statistically significant, while the diversity variables do not achieve statistical significance.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.3608875	*	0.203991	0.080	-2.0337	40.4229
Predicted Diversity	-3.019667		1.852362	0.106	-0.7241	8.3435
EF x Predicted Diversity	0.6667605	**	0.2853539	0.022	2.8191	51.2336
Constant	2.61055	*	1.326244	0.052		

Observations	100		
Overall R2	0.51422		
Root MSE	0.1651143		
F-stat. Model	33.87409	***	0.000
Log Likelihood	40.25899		

Table 4. “Basic” model for year 2005.

In both Table 3 and Table 4, the Rego method reveals that approximately 50% of the variation in income redistribution is attributed to the interaction between Economic Freedom and Predicted Diversity. This highlights the critical role of this interaction term in explaining income redistribution dynamics. It suggests that when assessing income inequality, the interplay between economic freedom and genetic diversity significantly influences the outcomes.

Furthermore, the Economic Freedom Index on its own demonstrates substantial explanatory power, accounting for approximately 40% of the R-squared in these models. This finding underscores the significance of economic freedom in shaping income redistribution. A higher degree of economic freedom is associated with lower income redistribution, aligning with the principles of capitalist economies.

In contrast, Predicted Diversity, while still contributing to the explanation of income redistribution, has a relatively lower descriptive value. It explains only around 13% of the variation in income redistribution in Table 3 and even less (8%) in the 2005 model (Table 4). This suggests that genetic diversity, as represented by Predicted Diversity, is not as influential in explaining income redistribution as economic freedom.

This pattern of results is consistent with other regression models discussed in subsequent sections. It highlights the enduring importance of the interaction between economic freedom and genetic diversity in shaping income inequality. These results imply that policies and factors related to economic freedom and genetic diversity play a central role in determining income redistribution outcomes.

Log GDP per capita

In this section, we delved into a series of econometric models, each introducing additional independent variables beyond the "basic" model discussed previously.

In the model presented in Table 5, the introduction of Log GDP per Capita (5-year average) as an independent variable brings about a significant shift in explanatory power. This variable takes center stage, contributing the largest share to the R-squared measure, approximately 35.8%. This substantial influence underscores the critical role played by a country's economic prosperity, as indicated by its GDP per capita, in elucidating variations in income redistribution.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.3144474	***	0.0510018	0.000	-1.8820	22.4941
Predicted Diversity	-1.557558	***	0.4573758	0.001	-0.3735	11.0327
EF x Predicted Diversity	0.4875497	***	0.0739508	0.000	2.2082	30.7080
Log GDP per capita	0.0943558	***	0.0072138	0.000	0.4913	35.7651
Constant	1.281686		0.3325471	0.000		
Observations	918					
Overall R2	0.54898					
Root MSE	0.1585782					
F-stat. Model	277.8281	***		0.000		
Log Likelihood	390.4252					

Table 5. Log GDP per capita.

The results strongly indicate that a nation's level of economic well-being is a primary driver in shaping income redistribution dynamics. Notably, the relationship suggests that as a country's

GDP per capita increases, there is a tendency towards adopting policies aimed at reducing income inequality. This finding underscores the importance of economic factors in influencing income redistribution, emphasizing that wealthier nations often implement measures to promote income equality within their societies.

In essence, the economic prosperity of a country emerges as a crucial determinant of income redistribution, highlighting the central role of economic well-being in fostering greater income equality. This result underscores the need for policymakers to consider economic factors when addressing income inequality and designing effective redistribution policies.

KOF economic globalization index

In the subsequent model, we introduced and evaluated the KOF economic globalization index (Flows) (5-year average) as an independent variable to gauge its impact on income redistribution. However, the results reveal that this variable possesses comparatively limited explanatory value, contributing only approximately 9.3% to the overall R-squared measure.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.4280454	***	0.0549081	0.000	-2.5783	32.9243
Predicted Diversity	-3.243505	***	0.4830141	0.000	-0.7828	12.7914
EF x Predicted Diversity	0.7444594	***	0.0788211	0.000	3.3971	44.9109
KOF globalization index	-0.0003385		0.0003505	0.334	-0.0297	9.3734
Constant	2.913054		0.3401836	0.000		
Observations	928					
Overall R2	0.46535					
Root MSE	0.1722121					
F-stat. Model	200.8434	***		0.000		
Log Likelihood	318.1103					

Table 6. KOF economic globalization index.

These findings [Table 6] imply that the KOF economic globalization index, which measures a nation's level of economic integration with the global economy, does not significantly account for variations in income redistribution among countries. This result may seem counterintuitive since globalization has often been associated with potential drivers of income inequality.

Several hypothetical reasons could explain this outcome. Firstly, it is possible that the influence of globalization on income redistribution is more complex and nuanced than previously assumed. While globalization may create economic disparities, countries may also implement

policies to mitigate these disparities, resulting in a less direct relationship between globalization and income redistribution.

Additionally, other factors such as domestic economic policies, political institutions, and cultural norms might exert a more substantial influence on income redistribution dynamics. These factors may overshadow the impact of globalization in explaining income redistribution variations across countries.

Continent-Specific Factors

In this model, depicted in Table 7, we introduced dummy variables representing the geographical regions where the explored countries are located: Europe, Asia, Africa, North America, and Oceania dummies. The results reveal interesting insights into the influence of regional factors on income redistribution. The North America dummy and Oceania dummy contribute relatively little to the explanation of income redistribution, with explanatory percentages of approximately 1.4% and 0.6%, respectively. This suggests that being located in North America or Oceania has a minor impact on a country's income redistribution policies.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.2139279	***	0.0535148	0.000	-1.2916	16.2121
Predicted Diversity	-2.472675	***	0.5498522	0.000	-0.5955	7.8347
EF x Predicted Diversity	0.3572763	***	0.076222	0.000	1.6327	21.5049
Europe	0.3876558	***	0.0447977	0.000	0.8009	37.5349
Asia	0.062319	*	0.0375592	0.097	0.1006	7.2245
Africa	0.1118312	**	0.0477413	0.019	0.1855	7.6483
North America	0.2389911	***	0.0358268	0.000	0.1666	1.4217
Oceania	0.1377527	***	0.0305718	0.000	0.1109	0.6188
Constant	2.524395		0.3694766	0.000		
Observations	943					
Overall R2	0.66009					
Root MSE	0.1376042					
F-stat. Model	226.7203	***		0.000		
Log Likelihood	536.7842					

Table 7. Continent-Specific Factors.

In contrast, the Europe dummy stands out with a substantial contribution, accounting for approximately 37.5% of the overall R-squared. This indicates that the region of Europe plays a significant role in shaping income redistribution outcomes. Interestingly, Europe's explanatory

weight appears to overshadow the contributions of other variables, such as the Economic Freedom Index and the interaction between Economic Freedom and Predicted Diversity.

Furthermore, Asia and Africa both contribute similarly to the explanation of income redistribution, with explanatory weights ranging from 7.2% to 7.6%. This is noteworthy considering that our model includes eight variables, and the relatively equal distribution of R-squared percentages among these regions suggests that both Asia and Africa have a meaningful impact on income redistribution patterns.

In summary, this model highlights the substantial influence of the European region on income redistribution, while also emphasizing the relevance of Asia and Africa. The low explanatory power of North America and Oceania implies that regional factors alone do not explain the variations in income redistribution, and a more comprehensive set of determinants is needed to capture the dynamics of income equality and redistribution across countries.

Legal Origins

The inclusion of variables representing the legal origin of explored countries in this model sheds light on the role of legal systems in explaining variations in income redistribution. Particularly, the prominence of "Legal Origin – Scandinavia" in explaining nearly 38% of the variation in income redistribution highlights the significance of Scandinavian legal heritage in fostering income equality [Table 8].

There are several hypothetical reasons for this outcome. One possibility is that Scandinavian legal systems are structured in a way that inherently promotes income redistribution, possibly through strong social safety nets, progressive taxation policies, or other mechanisms. This could be a reflection of Scandinavian countries' commitment to social welfare and equality-driven policies.

Another perspective is that the legal origin variable might serve as a proxy for a broader set of societal and institutional factors within Scandinavian countries that contribute to their success in addressing income inequality. These countries may have well-functioning political systems, strong labor unions, and a culture of social cooperation, all of which can facilitate income redistribution.

However, it is essential to remain cautious about attributing this effect solely to legal heritage. It is possible that legal origin is more emblematic of the geographical location and history of certain countries rather than a direct driver of income redistribution policies.

Regressor	Coefficient	Std.Err.	P>t	Std.Coeff.	Shapley %R2
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Economic Freedom	-0.228333	***	0.0445102	0.000	-1.3786	20.9818
Predicted Diversity	-1.697206	***	0.3860145	0.000	-0.4088	6.9298
EF x Predicted Diversity	0.4182109	***	0.0636514	0.000	1.9112	27.7008
Socialist	0.1014068	***	0.0141747	0.000	0.1464	2.8777
Germany	0.1387486	***	0.0218407	0.000	0.1311	3.7233
Scandinavia	0.4417192	***	0.0201912	0.000	0.4661	37.7866
Constant	1.943163		0.2702632	0.000		
Observations	943					
Overall R2	0.65123					
Root MSE	0.1392361					
F-stat. Model	291.2883	***		0.000		
Log Likelihood	524.658					

Table 8. Legal Origins.

In contrast to the substantial explanatory power of "Legal Origin – Scandinavia," the legal origins of socialism and Germany do not wield significant influence in explaining income redistribution dynamics. These variables contribute only 2.9% and 3.7%, respectively, to the overall R-squared in the income redistribution models.

Several hypothetical reasons could account for this divergence in explanatory power. For "Legal Origin – Socialist," it is possible that countries with a socialist legal heritage exhibit a more diverse range of income redistribution policies. Some socialist countries may have highly effective income redistribution mechanisms, while others may not. This variability could dilute the overall impact of the variable on the models. The same may hold true with the German legal origins.

Secondly, it could be that other contextual factors, such as economic conditions, political systems, or cultural attitudes, are more influential in shaping income redistribution in countries with these legal origins. These factors may overshadow the effects of legal heritage in this context.

Future research in this area could delve deeper into understanding the mechanisms through which legal systems influence income redistribution and whether there are specific policy approaches within Scandinavian countries that contribute to their effectiveness in addressing income inequality. Additionally, exploring the historical and contextual factors that have shaped legal origins and their impact on contemporary income redistribution policies could provide valuable insights into the complex relationship between legal systems and economic equality.

Religious Composition

In these following four models, we scrutinized the influence of the dominant religion's share within a country's population on income redistribution.

a) Share of Muslims

The findings reveal that the Share of Muslims in the population in 1980 plays a noteworthy role in describing variations in income redistribution, contributing approximately 10.5% to the overall R-squared [Table 9]. Intriguingly, this variable has a negative effect on income redistribution, suggesting that countries where Islam is the predominant religion tend to exhibit higher levels of income inequality compared to countries with other religious majorities.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.3608941	***	0.0535202	0.000	-2.1790	32.0937
Predicted Diversity	-2.342508	***	0.4779709	0.000	-0.5642	13.8171
EF x Predicted Diversity	0.6289909	***	0.0765387	0.000	2.8745	43.5810
Muslim	-0.0014196	***	0.0002329	0.000	-0.1666	10.5083
Constant	2.367924		0.3321368	0.000		
Observations	943					
Overall R2	0.48837					
Root MSE	0.1684599					
F-stat. Model	223.842	***		0.000		
Log Likelihood	343.9851					

Table 9. Share of Muslims.

Several hypothetical reasons could explain this phenomenon. Firstly, it is important to consider the socio-economic and cultural factors associated with predominantly Muslim countries. These nations may have unique historical, cultural, or institutional characteristics that influence income redistribution dynamics. Additionally, the role of Islamic law, or Sharia, in economic practices could affect income inequality, as it often emphasizes individual responsibility for charity and helping the less fortunate rather than state-driven income redistribution.

Furthermore, geopolitical factors and regional conflicts prevalent in some predominantly Muslim regions may contribute to income inequality. For instance, areas with prolonged conflicts or political instability may struggle to implement effective income redistribution policies. It is also essential to recognize that the Share of Muslims variable reflects not only religious identity but also regional and cultural diversity, making it a complex determinant of income redistribution.

b) Share of Protestants

The Share of Protestants in the population (1980) emerges as a substantial contributor, explaining nearly 30% of the overall R-squared in the models as Table 10 shows us. This suggests a strong association between a higher proportion of Protestants within a country's population and lower income inequality.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.4071476	***	0.0486055	0.000	-2.4671	26.1296
Predicted Diversity	-3.020907	***	0.4239532	0.000	-0.7292	9.6022
EF x Predicted Diversity	0.6797797	***	0.0690943	0.000	3.1160	34.9771
Protestant	0.0031901	***	0.0002232	0.000	0.3395	29.2912
Constant	2.84012		0.2980176	0.000		
Observations	935					
Overall R2	0.56292					
Root MSE	0.1557602					
F-stat. Model	299.4432	***		0.000		
Log Likelihood	414.3731					

Table 10. Share of Protestants.

Several hypotheses could shed light on these results. Firstly, the historical context of Protestantism, with its emphasis on individual responsibility and work ethic, may contribute to reduced income inequality. Protestant-majority countries may have developed economic and social systems that promote greater income equality through mechanisms like strong social safety nets, progressive taxation, and policies aimed at reducing wealth disparities.

Additionally, cultural factors associated with Protestantism, such as a strong sense of community and social cohesion might encourage income redistribution and support for policies that reduce inequality. Protestant-majority countries may prioritize social welfare and economic equality as part of their cultural values.

c) Share of Catholics

The examination of the impact of the predominant religion, specifically Catholicism, on income inequality reveals less substantial results. The Share of Catholics in the population (1980) contributes only 1% to the overall R-squared in the models, indicating a minimal role in explaining income inequality [Table 11].

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
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Economic Freedom	-0.3980978	***	0.0559223	0.000	-2.4036	36.9664
Predicted Diversity	-2.865214	***	0.5073096	0.000	-0.6901	13.1673
EF x Predicted Diversity	0.6970473	***	0.0795562	0.000	3.1855	48.8103
Catholic	0.0002663		0.0001745	0.127	0.0434	1.0559
Constant	2.637064		0.3585293	0.000		
Observations	943					
Overall R2	0.46943					
Root MSE	0.1715501					
F-stat. Model	207.4779	***		0.000		
Log Likelihood	326.8434					

Table 11. Share of Catholics.

These findings suggest that the proportion of the Catholic population in a given country may not significantly influence real-life income inequality dynamics. It is noteworthy that while the beta coefficient indicates a positive correlation between the presence of a larger Catholic population and income redistribution (improved income equality), the Rego method demonstrates that this correlation does not provide a substantial explanation for the observed data.

Several hypotheses can be proposed to contextualize these results. Firstly, the influence of religion on income inequality may be more nuanced and complex than a simple linear relationship. Other social, economic, and political factors may play more significant roles in shaping income redistribution policies and outcomes.

Additionally, the specific teachings and practices of the Catholic Church, as well as variations in interpretation and implementation across different countries, could contribute to the lack of a consistent relationship between Catholicism and income inequality. Moreover, historical and regional variations in the role of the Catholic Church in social and economic matters may also explain these findings.

d) Share of Other Religions

Similarly, when exploring the influence of other religions, the findings that are represented in Table 12, reveal a limited explanatory value. The Share of Other Religions in the population in 1980 contributes only 5.6% to the overall R-squared in the models. Interestingly, the correlation between this variable and income redistribution is negative, suggesting that countries with a higher prevalence of religions other than the major ones may experience greater income inequality.

However, despite this intriguing correlation, the Share of Other Religions variable does not provide a robust explanation for the observed data. The Rego method shows that its contribution to the overall R-squared is relatively low, indicating that other factors not captured in this variable are likely more influential in shaping income inequality patterns.

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.3526227	***	0.0532923	0.000	-2.1367	35.0659
Predicted Diversity	-2.394206	***	0.4701571	0.000	-0.5779	12.6968
EF x Predicted Diversity	0.6315036	***	0.0757173	0.000	2.8947	46.6664
Other religions	-0.0012798	***	0.000187	0.000	-0.1684	5.5708
Constant	2.359439		0.3290238	0.000		
Observations	935					
Overall R2	0.49249					
Root MSE	0.1678417					
F-stat. Model	225.6197	***		0.000		
Log Likelihood	344.5257					

Table 12. Share of Other Religions.

These results prompt further exploration and hypotheses. It is possible that the negative correlation between the prevalence of other religions and income redistribution is indicative of complex social and economic dynamics. Societies with greater religious diversity might face unique challenges or opportunities that affect income inequality, but these dynamics are not fully captured by this variable alone.

Future research could delve into the specific religious compositions within the category of "other religions" and examine how these diverse religious groups interact with socioeconomic and political factors to influence income inequality. Additionally, qualitative research methods may help uncover nuanced insights into the role of religion in income redistribution, moving beyond simple quantitative associations.

6. Combination of Factors

In this section, we extend our analysis by examining the combined impact of various factors in three additional regression models. The results reveal compelling insights into the explanatory power of these variables.

“Basic” model, Including Economic Indicators and Continent Dummies

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.2262835	***	0.0542701	0.000	-1.3506	12.0265
Predicted Diversity EF x Predicted Diversity	-2.359705	***	0.5560253	0.000	-0.5667	7.1595
Log GDP per capita	0.3686892	***	0.0775661	0.000	1.6667	16.6856
KOF globalization index	0.0347676	***	0.0082534	0.000	0.1811	16.2377
Europe	-0.0011908	***	0.0002997	0.000	-0.1047	4.1064
Asia	0.3404709	***	0.0461745	0.000	0.7038	30.6772
Africa	0.0367451		0.0385046	0.340	0.0588	6.1004
North America	0.1154759	**	0.0487321	0.018	0.1920	5.4641
Oceania	0.1797478	***	0.0378343	0.000	0.1268	1.0072
Constant	0.1230135	***	0.0323147	0.000	0.0945	0.5353
Observations	2.256734		0.3825278	0.000		
Overall R2	915					
Root MSE	0.66658					
F-stat. Model	0.1367984					
Log Likelihood	180.7331	***		0.000		

Table 13. “Basic” model, Including Economic Indicators and Continent Dummies.

In the first model, which combines the "basic" model, economic factors, and Continent Dummies, we find consistent patterns with previous findings. Economic Freedom (EF) continues to play a substantial role, contributing 12% to the overall R-squared. The interaction term EF*PD remains a significant factor, along with Log GDP per capita, each explaining approximately 16% of the variance in income redistribution. Notably, the Europe dummy exerts a substantial influence, accounting for 30% of the overall explanatory power. Conversely, Oceania's contribution is marginal, comprising only 0.5% of the total R-squared.

The Europe dummy's prominent role suggests that regional economic and political factors play a crucial role in shaping income redistribution outcomes, possibly due to the European Union's extensive social welfare policies. Conversely, Oceania's limited impact on income redistribution could be attributed to the smaller number of countries in this region or variations in their policy approaches.

“Basic” model, Including Economic Indicators and Legal Origins

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
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Economic Freedom	-0.1765359	***	0.0432155	0.000	-1.0537	13.1687
Predicted Diversity	-0.6044786		0.3885108	0.120	-0.1452	6.8590
EF x Predicted Diversity	0.274998	***	0.0634491	0.000	1.2431	18.3939
Log GDP per capita	0.0812612	***	0.0062781	0.000	0.4232	22.2101
KOF globalization index	-0.0003979		0.0002857	0.164	-0.0350	4.4328
Socialist	0.0484179	***	0.0139781	0.001	0.0699	1.4586
Germany	0.0926004	***	0.021518	0.000	0.0885	2.5850
Scandinavia	0.4060038	***	0.0191937	0.000	0.4301	30.8919
Constant	0.7898674		0.2798431	0.005		
Observations	915					
Overall R2	0.70094					
Root MSE	0.129415					
F-stat. Model	265.4415	***		0.000		
Log Likelihood	577.1224					

Table 14. “Basic” model, Including Economic Indicators and Legal Origins.

In the second model, we again combine the "basic" model, economic factors, and Continent Dummies, yielding results in line with our prior analyses. EF maintains a substantial presence, contributing 13% to the R-squared. The interaction term EF*PD, alongside Log GDP per capita, significantly explains variations in income redistribution, constituting approximately 18% and 22% of the R-squared, respectively. Interestingly, the presence of the Scandinavian legal origin is pivotal in this model, contributing a substantial 30% to the overall R-squared. In contrast, the Socialist legal origin has minimal influence, contributing only 1.5% to the R-squared.

The prominence of the Scandinavian legal origin in this model may reflect the strong welfare states and income redistribution policies found in Scandinavian countries. In contrast, the limited influence of the Socialist legal origin could be due to variations in socialist economic models and approaches to income redistribution.

“Basic” model, Including Economic Indicators and Religious Composition

Regressor	Coefficient		Std.Err.	P>t	Std.Coeff.	Shapley %R2
Economic Freedom	-0.199264	***	0.0453717	0.000	-1.1936	13.4531
Predicted Diversity	-0.1389207		0.4239037	0.743	-0.0334	9.3862
EF x Predicted Diversity	0.2876806	***	0.0664182	0.000	1.3044	19.1128
Log GDP per capita	0.1010018	***	0.0062139	0.000	0.5275	24.7485
KOF globalization index	-0.0010462	***	0.0002826	0.000	-0.0922	4.3001
Protestant	0		0	.	0.0000	17.6344

Muslim	-0.003971	***	0.0002518	0.000	-0.4641	5.2721
Catholic	-0.0027442	***	0.0002073	0.000	-0.4457	2.3194
Other religions	-0.0040927	***	0.0002245	0.000	-0.5358	3.7735
Constant	0.7414772	**	0.3029046	0.015		
Observations	907					
Overall R2	0.68204					
Root MSE	0.1335107					
F-stat. Model	240.7834	***		0.000		
Log Likelihood	543.8566					

Table 15. "Basic" model, Including Economic Indicators and Religious Composition.

The third model, merging the "basic" model, economic factors, and Continent Dummies, mirrors previous outcomes. EF remains influential, contributing 13.5% to the R-squared. The interaction term EF*PD, along with Log GDP per capita, accounts for significant portions of the income redistribution variance, representing approximately 19% and 24.7%, respectively. However, an intriguing observation emerges here: the Protestant regressor was omitted, yet it played a substantial role in explaining the goodness-of-fit, with a Shapley percentage of 17.6% of the R-squared. Additionally, it is worth noting that the Catholic dummy experienced a change in its beta coefficient, becoming negative, but it still holds a modest 2% share in the goodness-of-fit measure.

The omission of the Protestant regressor, despite its substantial Shapley percentage, suggests potential multicollinearity issues or the need for a more nuanced analysis of religious factors in future research. The change in the beta coefficient for the Catholic dummy, along with its relatively small share in the goodness-of-fit measure, could indicate complex interactions between religious composition and income redistribution, warranting further investigation.

7. Conclusion

In conclusion, our comprehensive analysis of Ibrahim Ben Araar's research, building upon the foundation laid by Sturm & De Haan (2015) and Ashraf & Galor (2013), coupled with the application of the Goodness-of-fit decomposition method, has provided us with significant insights into the intricate world of income redistribution dynamics. These findings offer valuable implications for our understanding of economic equality and its potential real-world implications.

First and foremost, the profound impact of the interaction between economic freedom and predicted diversity on income redistribution is noteworthy. This interaction consistently emerges as a dominant explanatory factor across all regressions, illuminating the crucial role it plays in

shaping economic equity. The positive influence of this interaction suggests that countries striking a balance between economic freedom and diversity tend to exhibit more favorable income redistribution dynamics.

Conversely, the Economic Freedom Index, despite its strong explanatory power, exerts a negative effect on income redistribution. This intriguing finding underscores the intricate interplay between economic freedom and equitable income distribution. A plausible interpretation could be that an overly liberal economic environment, characterized by minimal government intervention, might inadvertently hinder efforts to mitigate income inequality.

Predicted diversity, while contributing less to the explanation of the dependent variable, consistently wields a potent negative effect on income redistribution. This observation raises questions about the complexities surrounding diversity's impact on economic equality. For instance, countries with high levels of predicted diversity might face challenges in achieving income redistribution due to potential social and economic disparities among various ethnic or cultural groups.

The noteworthy contribution of Log GDP per capita to the explanatory power of our models cannot be overlooked. It exerts a substantial positive influence on income redistribution, albeit with a slight impact. This implies that countries with higher average income levels tend to experience somewhat more equitable income distribution. This result aligns with the notion that greater prosperity often correlates with reduced income inequality, as resources are more widely distributed among the population.

On the other hand, the KOF Globalization Index appears to have a negligible negative effect on income redistribution, coupled with low explanatory value. This suggests that the globalization index does not significantly contribute to income inequality within individual countries. It is possible that the globalization index may capture a broad range of economic activities and interactions between nations, but its impact on income distribution at the national level remains subtle and inconclusive.

We assessed the North America and Oceania dummy variables, representing countries in these regions. Surprisingly, they showed low explanatory power, suggesting minimal influence on income redistribution despite a positive correlation. Possible reasons include differences in economic structures, social policies, or historical contexts.

Conversely, the Europe dummy had a significant positive effect on income redistribution, indicating that European countries tend to prioritize income equality through policies like social

welfare programs and progressive taxation. These results underscore the importance of considering regional differences in income redistribution for policymakers seeking effective strategies.

Asia and Africa exhibited moderate explanatory power with a slight positive correlation, indicating a role in shaping income inequality dynamics. Factors such as economic growth, political stability, and social programs likely contribute. Notably, significant variation exists in income redistribution practices within these continents among individual countries.

Our investigation into legal origins revealed striking differences among various categories. Legal origin – Scandinavia emerged as a dominant player, explaining a substantial share of redistribution volatility and displaying a considerable positive correlation with income redistribution. This suggests that countries with a legal heritage stemming from Scandinavia are particularly adept at addressing income inequality, either due to the legacy of their legal systems or other effective mechanisms in place. In contrast, legal origin – socialist and legal origin – Germany, while still exhibiting positive correlations, do not contribute significantly to the descriptive value of our models. This implies that their legal origins play a lesser role in shaping income redistribution dynamics, and other factors may be more influential in these cases.

In our analysis of religious compositions, we have observed distinct patterns. A higher proportion of Protestants is associated with slightly greater income equality. However, the omission of the Protestant regressor in the model in Table 15 highlights complex interactions in this relationship. Conversely, a larger Muslim population tends to correlate with higher income inequality, influenced by various socio-economic, political, and cultural factors. The contributions of other religions and Catholics are relatively minor, suggesting that religious composition has limited direct impact on income redistribution, with other factors taking precedence.

Despite the valuable insights gained, our analysis has certain **limitations**. Firstly, our study is based on aggregated data, which may overlook finer nuances at the individual level. Secondly, the causal relationships between variables remain unexplored, as our approach is primarily correlational. Additionally, the cultural and historical context of legal origins and religious compositions could provide a more nuanced understanding, but this is beyond the scope of our analysis. Lastly, our study focuses on selected factors, and there may be other unaccounted variables influencing income redistribution. These limitations call for caution in generalizing our findings and suggest avenues for more comprehensive future research.

Future research endeavors could consider addressing the limitations of our study and expanding on our findings. Investigating causal relationships between factors influencing income redistribution would deepen our understanding. Furthermore, delving into the historical and cultural contexts of legal origins and religious compositions could provide valuable insights. Exploring additional variables and conducting finer-grained analyses at the individual level may also yield a more comprehensive picture of income redistribution dynamics.

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Appendix

Table A 1: Variable List

Abbreviation	Variable description	Source
Redistribution	This variable represents the ratio between the market Gini coefficient and the net Gini coefficient. It measures the extent to which income inequality is reduced through government interventions or social programs.	Sturm & de Haan (2015)
Economic Freedom	This is a 5-year average of the Economic Freedom (EF) index, which assesses the level of economic freedom in a country, considering factors like property rights, rule of law, and government intervention in the economy etc. It is a proxy for capitalism.	Sturm & de Haan (2015)
Predicted Diversity	This variable represents Predicted Heterozygosity, which is calculated using migratory distance to East Africa. It provides insights into the genetic diversity of a population.	Ashraf and Galor (2013)
Log GDP per Capita	This is the logarithm of Gross Domestic Product (GDP) per capita, averaged over five years. It provides a measure of a country's economic prosperity. It is used in a model to account for and rectify any distributional effects influenced by income levels. Consistent with the Kuznets hypothesis, we anticipate that inequality will diminish as development levels rise, based on previous studies.	The World Bank (2015)
KOF Globalization Index	The KOF Economic Globalization Index (Flows) encompasses two key components. The first part is grounded in empirical cross-border flow data, encompassing trade, foreign direct investment, portfolio investment, and income payments. In contrast, the second part delves into trade restrictions, such as the presence of concealed import barriers, tariff rates, trade-related taxes, and capital account restrictions. Since this latter component fundamentally overlaps with the Economic Freedom index, we exclusively consider the segment of the KOF economic globalization index that pertains to actual economic flows.	Sturm & de Haan (2015)
Europe; Asia; North America; Oceania	These are dummy variables representing geographic regions. They take the value of 1 if the country is in that region and 0 otherwise.	Sturm & de Haan (2015)

Socialist; Germany; Scandinavia	These are dummy variables representing legal origins or specific countries. They take the value of 1 if the country falls into the respective category and 0 otherwise.	Sturm & de Haan (2015)
Muslim; Protestant; Catholic; Other Religion	These variables represent the share of different religious groups in the population as of 1980. They measure the religious composition of the population.	Sturm & de Haan (2015)

Table A 2: Descriptive Statistics

Label	Definitions	Obs.	Mean	Std.Dev.	Min	Max
Redistribution	Ratio between market and net Gini coefficient	4563	1.175	.210	.673	2.161
Economic Freedom	5— Year Average of Economic Freedom Index	1664	6.247	1.424	2.016	9.020
Predicted Diversity	Predicted Heterozygosity using migratory distance to East — Africa	19177	.695	.059	.572	.774
Log GDP per capita	Log GDP per capita (5 — year average)	8274	8.315	1.223	5.031	11.734
KOF globalization index	KOF economic globalization index (Flows) (5 — year average)	6786	51.568	21.885	3.140	100
Europe	Europe dummy	19178	.289	.453	0	1
Asia	Asia dummy	19178	.200	.400	0	1
North America	North America dummy	19178	.023	.151	0	1
Oceania	Oceania dummy	19178	.059	.235	0	1

Socialist	Legal origin — socialist	19032	.151	.358	0	1
Germany	Legal origin - Germany	19032	.048	.214	0	1
Scandinavia	Legal origin — Scandinavia	19032	.044	.205	0	1
Muslim	Share of Muslims in population (1980)	19042	18.174	33.558	0	99.9
Catholic	Share of protestants in population (1980)	19042	38.275	38.757	0	99.1
Protestant	Share of Catholics in population (1980)	18811	14.384	23.646	0	99.8
Other religions	Share of other religions	18811	29.004	32.258	0	100
